Assignment 3

Deadline: Monday, 2 November 2026, 18:00

Introduction

In this assignment, you will implement three building blocks for approximate inference in Bayesian networks: *topological sorting*, *forward sampling*, and *rejection sampling*.

The provided archive `assignment3.zip` (link) contains all files needed for this assignment:

- `rejection_sampling.ipynb`: the notebook in which you complete the programming tasks.
- `BayesNet_class_description.ipynb`: an introduction to the provided Bayesian network class and its usage.
- `bayesian_network.py` and `utils.py`: supporting Python modules used by the notebooks.
- `environment.yml`: a specification of the required Python environment; use `conda env create -f environment.ym` to install the environment and `conda activate pm` to activate it.

Submission instructions:

- Submit your completed **Jupyter notebook (`.ipynb`)** via Moodle. Submit **your Jupyter notebook only**; do not upload supplementary files or a slide deck with detailed explanations. You are not required to create solution slides.

Do not forget to:

- indicate which parts of the assignment you have completed here;
- select an interview time slot here;
- provide feedback here.

1 Topological sort (2 points)

Implement the `__iter__` method for the `BayesNet` class in the Jupyter notebook.

2 Forward sampling (1 point)

Implement `sample_forward(bayes_net, sample_size)`.

3 Rejection sampling (1 point)

Implement `rejection_sampling(bayes_net, ...)`.